

Questions

1. Explain the difference between object and class!

Ans :

A class is a blueprint or template used to create objects, while an object is a specific instance created from a class. For example, BikeDemo is a class that defines the properties and behaviors of a bike, while mountainBike1 is an object created from the BikeDemo class.

2. State your reason why gear and brand can be classified as attribute for Bike object!

Ans :

Gear and brand can be classified as attributes of a Bike object because they describe the characteristics or properties of the bike. Brand stores information about the bike manufacturer, while gear stores the current gear of the bike.

3.State one of OOP better point than procedural programming!

Ans:

One advantage of OOP over procedural programming is that OOP organizes programs into classes and objects, making the code easier to organize, reuse, maintain, and modify.

4.Is it allowed to define two attributes in one line code such “public String name, address;”?

Ans:

Yes, it is allowed to define two attributes in one line, such as public String name, address;. Both name and address will be declared as String attributes. However, writing them on separate lines can make the code easier to read

5. In RoadBike class, state your reason why brand, speed, and gear attributes are not written again in this class!

Ans :

The brand, speed, and gear attributes are not written again in the RoadBike class because RoadBike inherits them from the BikeDemo class. Through inheritance, RoadBike can use the attributes and methods of BikeDemo without having to declare them again.

Result

PROBLEMS 1	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS
<pre>ashfaq-khan@ashfaq-khan-Nitro-AN515-58:/run/media/ashfaq-khan/visual Studio/All projects/object programing\$ /usr/bin/env /usr/lib/jvm/java-21-openjdk-amd64/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /home/ashfaq-khan/.config/Code/User/workspaceStorage/02c7fee62990b8f1c89dd9208c0423a2/redhat.java/jdt_ws/object\ programing_149bec98/bin objectprogramming.jobsheet1.expl.Main Gear : 3 ashfaq-khan@ashfaq-khan-Nitro-AN515-58:/run/media/ashfaq-khan/visual Studio/All projects/object\ programing\$ cd /run/media/ashfaq-khan/visual Studio/All projects/object\ programing ; /usr/bin/env /usr/lib/jvm/java-21-openjdk-amd64/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /home/ashfaq-khan/.config/Code/User/workspaceStorage/02c7fee62990b8f1c89dd9208c0423a2/redhat.java/jdt_ws/object\ programing_149bec98/bin objectprogramming.jobsheet1.expl.Main Brand : Trek Speed : 5 Gear : 2 Brand : Gaint Speed : 10 Gear : 3 Brand : Speciaalized Speed : 5 Gear : 4 Tire Width : 25mm Bike Type : Road Bike ashfaq-khan@ashfaq-khan-Nitro-AN515-58:/run/media/ashfaq-khan/visual Studio/All projects/object\ programing\$</pre>				